Memory Forensics Report

Case MT-20261022-003706-941B | Generated 2026-10-21 21:38:04 UTC

1. Executive Summary

Compromised | Risk Score 100/100

CRITICAL	HIGH	MEDIUM	LOW	INFO
1	2	0	2	0

MemTrace analyzed the memory image **0zapftis.vmem** and identified **5** finding(s) across 22 recovered processes, 0 network artifacts and 12 suspicious memory regions. The most significant findings are listed below and detailed in Section 3.

ID	Severity	Finding
F-001	CRITICAL	AppInit DLL mfc42ul.dll loaded into 15 processes
F-002	HIGH	AppInit_DLLs registry value configured: mfc42ul.dll
F-003	HIGH	Look-alike DLL mfc42ul.dll resembles mfc42u.dll

2. Evidence and Methodology

Evidence Item

File Name	0zapftis.vmem
Size	256.0 MB (268,435,456 bytes)
SHA-256	`eb8cc1f46cf0ed2ada1b467916a684ee8ab3e5ce2cb66ee581f89d921e4e539b`
MD5	cb47af710c7ab59e43d1396aa7ce950c
Integrity	Verified - hash unchanged after analysis

Analysis Environment

Tool	MemTrace 1.0.0
Analysis Engine	Volatility 2.28.2
Python	3.13.1
Analyst OS	Windows 11
Started (UTC)	2026-10-21T21:37:06+00:00
Finished (UTC)	2026-10-21T21:38:03+00:00

Target System Profile

64-bit	False
PAE	True
NT Major Version	5
NT Minor Version	1
Product Type	NtProductWinNt
System Root	C:\WINDOWS

System Time at Capture	2011-10-10 17:06:54+00:00
Build Lab	2600.xpsp_sp2_rtm.040803-2158
Processors	1

Methodology

The evidence was hashed before and after analysis to demonstrate that it was not modified. Volatility 3 plugins were executed against the image to enumerate processes from the active process list (pslist) and by pool scanning (psscan), reconstruct the process tree and command lines, list loaded modules (dlllist), read the AppInit_DLLs registry value from the in-memory registry (printkey), recover network artifacts (netscan) and locate private executable memory regions (malfind). MemTrace detection rules then compared these sources to identify hidden processes, masquerading processes and modules, abnormal parent-child relationships, DLL injection persistence, suspicious network activity and code injection.

Plugin	Status	Detail	Time (s)
windows.info	OK	23 rows	3.9
windows.pslist	OK	22 rows	3.1
windows.psscan	OK	22 rows	5.5
windows.pstree	OK	22 rows	3.0
windows.cmdline	OK	22 rows	3.7
windows.dlllist	OK	847 rows	6.7
windows.netscan	FAILED	windows.netscan failed: NotImplementedError: This version of Windows is not supported: 5.1 15.2600!	4.9
windows.malware.malfind	OK	12 rows	13.0
windows.registry.printkey --key "Microsoft\Windows NT\CurrentVersion\Windows"	OK	19 rows	5.1
windows.registry.printkey --key "Wow6432Node\Microsoft\Windows NT\CurrentVersion\Windows"	OK	13 rows	4.0

3. Detailed Findings

CRITICAL F-001 AppInit DLL mfc42ul.dll loaded into 15 processes

The DLL configured in AppInit_DLLs is mapped into running processes, so its code is executing inside them. Malware uses this to run in every interactive and service process at once.

Category	Injection
Rule	APPINIT_DLL_LOADED
Process	mfc42ul.dll
MITRE ATT&CK	T1546.010, T1055.001
Path	`C:\WINDOWS\System32\mfc42ul.dll, C:\WINDOWS\system32\mfc42ul.dll`
Process Count	15
PIDs	`184, 192, 228, 400, 544, 632, 676, 688, 832, 848, 964, 1260, 1444, 1920, 1956`

Recommendation: Isolate the host, dump the DLL from memory for analysis and identify how the registry value was set.

HIGH F-002 AppInit_DLLs registry value configured: mfc42ul.dll

The AppInit_DLLs registry value forces Windows to load the listed DLLs into every process that loads user32.dll. It is rarely used by legitimate software and is a well-known malware persistence and injection technique.

Category	Persistence
Rule	APPINIT_DLLS
Process	mfc42ul.dll
MITRE ATT&CK	T1546.010
Registry Key	`\Device\HarddiskVolume1\WINDOWS\system32\config\software\Microsoft\Windows NT\CurrentVersion\Windows`
Value	mfc42ul.dll
LoadAppInit_DLLs	not present
Last Write Time	2011-10-10T16:56:35+00:00

Recommendation: Acquire the listed DLLs from disk, verify them against known-good hashes and remove the registry value.

HIGH F-003 Look-alike DLL mfc42ul.dll resembles mfc42u.dll

A loaded module named 'mfc42ul.dll' is a near-copy of the legitimate Windows library 'mfc42u.dll'. Attackers give malicious DLLs look-alike names so they blend in with system libraries.

Category	Module
Rule	DLL_MASQUERADING
Process	mfc42ul.dll
MITRE ATT&CK	T1036.005
Resembles	mfc42u.dll
Path	`C:\WINDOWS\System32\mfc42ul.dll, C:\WINDOWS\system32\mfc42ul.dll`
Process Count	15
PIDs	`184, 192, 228, 400, 544, 632, 676, 688, 832, 848, 964, 1260, 1444, 1920, 1956`

Recommendation: Acquire the DLL, compare it with the legitimate library and check its digital signature.

LOW F-004 Low-confidence executable memory in csrss.exe (PID 608)

Executable regions were flagged by malfind but start with zero bytes or a Windows heap header, so no code is present at the region entry. These are common on older Windows versions and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	csrss.exe (PID 608)
MITRE ATT&CK	T1055
Regions	1
Empty regions	0
Heap regions	1
Addresses	0x7f6f0000

Recommendation: No action needed unless other findings involve this process.

LOW F-005 Low-confidence executable memory in winlogon.exe (PID 632)

Executable regions were flagged by malfind but start with zero bytes or a Windows heap header, so no code is present at the region entry. These are common on older Windows versions and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	winlogon.exe (PID 632)
MITRE ATT&CK	T1055
Regions	11
Empty regions	9
Heap regions	2
Addresses	`0x580000, 0x64f0000, 0x4e5d0000, 0x23df0000, 0x2b300000, 0x4bd80000, 0x51490000, 0x66450000`

Recommendation: No action needed unless other findings involve this process.

4. Process Listing

PID	PPID	Name	Created	Source	Flags
4	0	System		pslist	
184	1956	VMwareTray.exe	2011-10-10T17:04:41	pslist	appinit-dll, masquerading-dll
192	1956	VMwareUser.exe	2011-10-10T17:04:41	pslist	appinit-dll, masquerading-dll
228	1956	reader_sl.exe	2011-10-10T17:04:41	pslist	appinit-dll, masquerading-dll
400	964	wuauclt.exe	2011-10-10T17:04:46	pslist	appinit-dll, masquerading-dll
536	4	smss.exe	2011-10-10T17:03:56	pslist	
544	1956	cmd.exe	2011-10-10T17:06:42	pslist	appinit-dll, masquerading-dll
608	536	csrss.exe	2011-10-10T17:03:58	pslist	
632	536	winlogon.exe	2011-10-10T17:03:58	pslist	appinit-dll, masquerading-dll
676	632	services.exe	2011-10-10T17:03:58	pslist	appinit-dll, masquerading-dll
688	632	lsass.exe	2011-10-10T17:03:58	pslist	appinit-dll, masquerading-dll
832	676	vmacthlp.exe	2011-10-10T17:03:59	pslist	appinit-dll, masquerading-dll
848	676	svchost.exe	2011-10-10T17:03:59	pslist	appinit-dll, masquerading-dll
916	676	svchost.exe	2011-10-10T17:03:59	pslist	
964	676	svchost.exe	2011-10-10T17:03:59	pslist	appinit-dll, masquerading-dll
1020	676	svchost.exe	2011-10-10T17:03:59	pslist	
1148	676	svchost.exe	2011-10-10T17:04:00	pslist	
1260	676	spoolsv.exe	2011-10-10T17:04:00	pslist	appinit-dll, masquerading-dll
1444	676	VMwareService.e	2011-10-10T17:04:00	pslist	appinit-dll, masquerading-dll
1616	676	alg.exe	2011-10-10T17:04:01	pslist	
1920	964	wscntfy.exe	2011-10-10T17:04:39	pslist	appinit-dll, masquerading-dll
1956	1884	explorer.exe	2011-10-10T17:04:39	pslist	appinit-dll, masquerading-dll

5. Network Artifacts

Network artifacts could not be recovered for this image (windows.netscan is not supported for this Windows version).

6. Limitations

This report was produced by automated analysis. Findings are investigative leads that should be validated by a qualified examiner. Absence of findings does not prove the system was not compromised; memory captures only a point-in-time state and advanced malware may evade these detection rules.